Sentiment-Driven Analysis of Cryptocurrency Trader Performance and Segmentation

Project Overview

This project investigates how cryptocurrency trader behavior and profitability are shaped by market sentiment as measured by the Fear & Greed Index. Using two datasets — historical trade records and sentiment classifications — the analysis spans data preprocessing, feature engineering, performance analysis, trader segmentation, and predictive modelling.

Methodology

Data Sources

- Fear & Greed Index dataset — daily sentiment classifications (Fear, Extreme Fear, Neutral, Greed, Extreme Greed)
- Historical trades dataset — individual trade records including account, size, PnL, side (Buy/Sell), and timestamp

Analytical Pipeline

- 1. Data Loading & Preprocessing:
  - Merged trade and sentiment datasets on date
  - Handled null values, type conversion, and deduplication
- 2. Feature Engineering:
  - Daily PnL aggregation per trader
  - Win/Loss flag, trade frequency count, average trade size
- 3. Performance Analysis:
  - PnL distribution and win rates grouped by sentiment
  - Trade volume and size patterns across market states
- 4. Trader Segmentation:
  - High vs. Low trade size (median split)
  - Consistent vs. Inconsistent (PnL standard deviation)
  - Frequent vs. Infrequent (trade count median split)
  - Profitable vs. Non-profitable (net PnL sign)

- 5. Predictive Modelling:
  - Random Forest Classifier trained on daily-level features
  - Target: next-day profitability (binary classification)

Key Findings at a Glance

Finding	Details
Sentiment & PnL	Greed: avg PnL 87.89 | Fear: 50.05 | Extreme Greed: 25.42 | Neutral: 22.23
Win Rate by Sentiment	Extreme Greed: 49.01% | Fear: 41.51% | Neutral: 31.72%
Trade Volume	Fear: 133,871 trades | Greed: 36,289 trades (Fear drives highest activity)
Avg Trade Size	Extreme Greed: $5,660 | Neutral: $3,059 (largest sizes in bullish extremes)
Buy/Sell Ratio	Slight sell bias in Greed (57.5%) and Extreme Greed (51.6%)
High vs Low Size	High-size traders earn 10–14× more PnL than low-size in Greed periods
Consistent vs Inconsistent	Inconsistent traders outperform in Fear, Greed, Neutral; Consistent better in Extreme Greed
Frequent vs Infrequent	Infrequent traders show higher PnL in Fear (75.64 vs 46.01), Greed (234.89 vs 70.58), Neutral (356.72 vs 4.06)
Predictive Model Accuracy	Random Forest: ~64.29% accuracy — better than random, confirms sentiment has predictive signal

Visualizations & Analysis

The following charts were generated during the analysis. Each visualization is accompanied by an interpretation of the key patterns and insights it reveals.

Figure 1: PnL Distribution by Market Sentiment

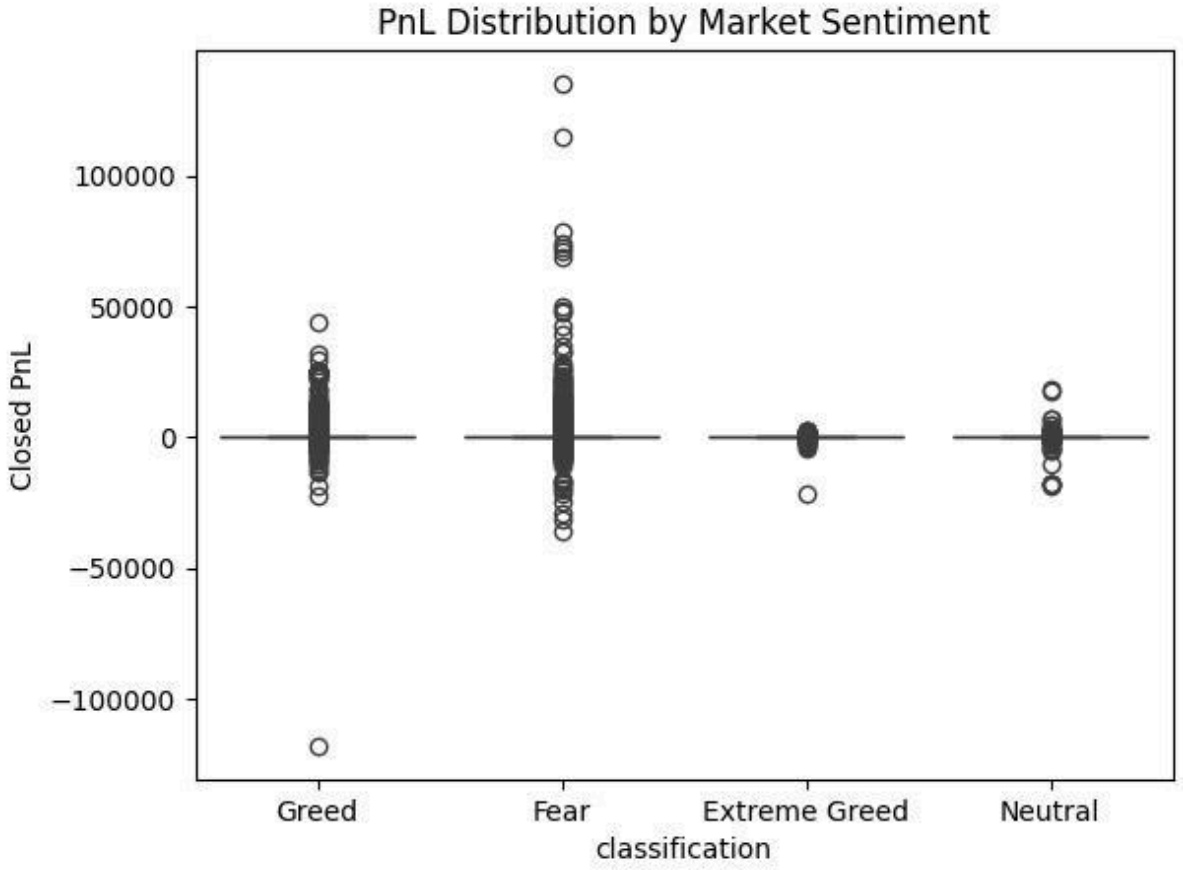

Figure 1 — Box/strip plot of Closed PnL across Greed, Fear, Extreme Greed, and Neutral sentiment classifications.

Interpretation

This strip plot reveals the full distribution of individual trade PnL values across each sentiment category, rather than just the average. Several critical patterns emerge:

- Wide outlier spread in Fear and Greed: Both categories contain extreme outliers reaching $100,000+ in gains and losses, indicating that highly volatile trades occur disproportionately in these sentiment states.
- Compressed distribution in Extreme Greed: Trades during Extreme Greed show a much tighter PnL cluster near zero, with fewer dramatic wins or losses, suggesting more cautious or smaller-scale trading behaviour.
- Neutral market volatility: Neutral sentiment shows a modest spread, but a cluster of negative outliers around -$10,000 to -$20,000 stands out, implying that directionless markets can still lead to meaningful losses.
- Central tendency near zero: Across all sentiment states, the median PnL sits close to zero, confirming that most individual trades are marginally profitable or unprofitable — large gains are the exception, not the norm.

Key Insight: *Fear markets, despite lower average win rates, host the most extreme individual trade outcomes. A small number of high-conviction trades during Fear may be responsible for the elevated average PnL in that category*

Figure 2: Number of Trades by Sentiment

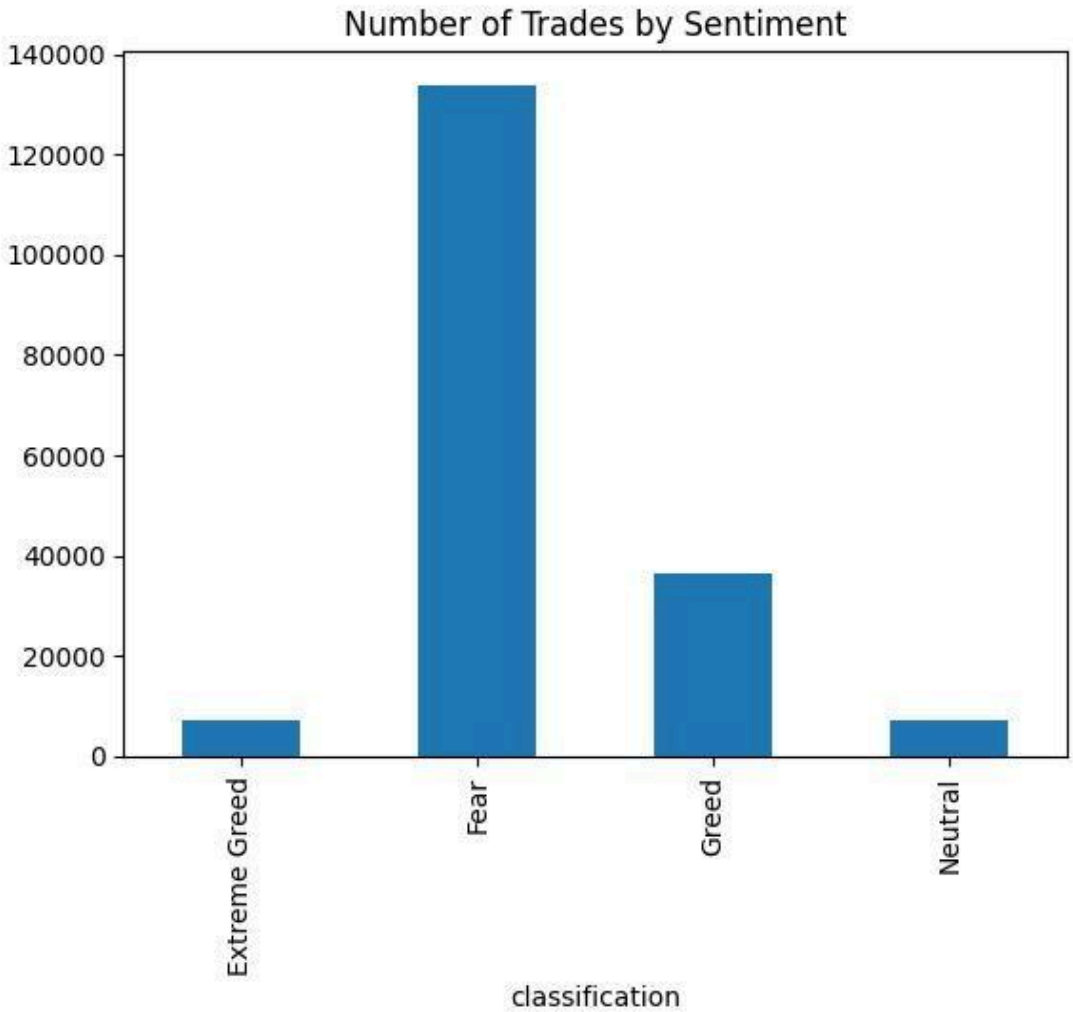

Figure 2 — Bar chart showing total trade count across Extreme Greed, Fear, Greed, and Neutral sentiment periods.

Interpretation

This bar chart makes the most dramatic finding in the dataset immediately visible — Fear-driven markets generate an order of magnitude more trading activity than any other sentiment state:

- Fear dominates trade volume at 133,871 trades — roughly 3.7× more than Greed (36,289) and 17× more than Extreme Greed or Neutral (~7,000–8,000 each).
- This spike in Fear-period activity suggests one of two behavioural patterns: panic selling (traders rushing to close losing positions) or opportunistic buying (traders sensing undervalued assets).
- The relatively low trade counts during Extreme Greed and Neutral suggest that traders are either more selective or less active during these sentiment periods.

- The data coverage imbalance (Fear periods dominate the dataset) is an important consideration when interpreting average PnL values — Fear's average PnL is weighted by a far larger number of trades.

Key Insight: *The overwhelming trade volume during Fear periods suggests that market panic is the primary driver of trading frequency in crypto markets — a finding with significant implications for liquidity, slippage, and execution quality analysis.*

Figure 3: Average Trade Size by Sentiment

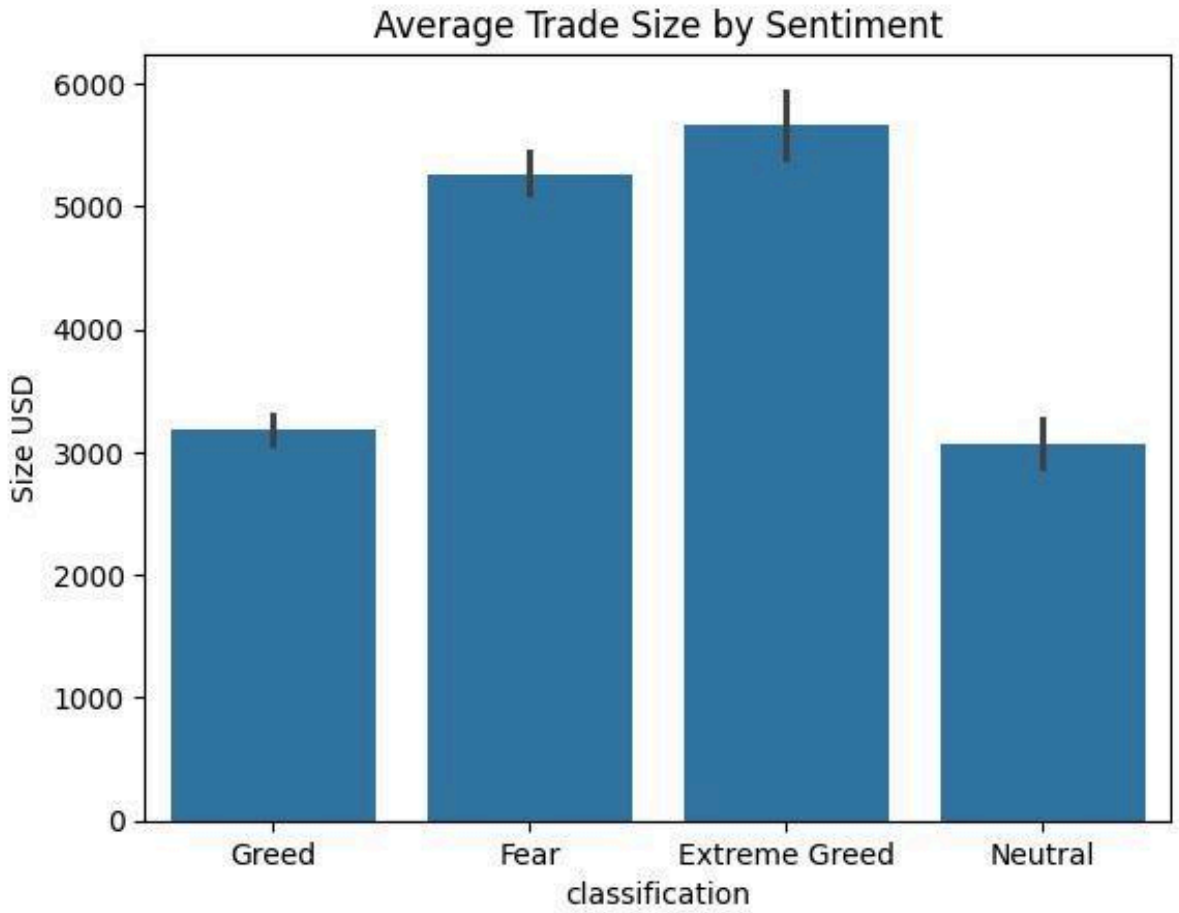

Figure 3 — Bar chart with confidence intervals showing average trade size (USD) across sentiment classifications.

Interpretation

Average trade size reveals how much capital traders are willing to commit under different market conditions, offering a window into risk appetite:

- Extreme Greed attracts the largest average trade sizes (~$5,660), indicating that traders are most willing to deploy capital when sentiment is at its most euphoric.
- Fear-period trades are also larger than average (~$5,280), suggesting that traders who do trade during fear periods commit meaningful position sizes — possibly reflecting high-conviction contrarian bets.

- Greed periods show surprisingly modest average trade sizes (~$3,200), close to the Neutral baseline (~$3,060), despite generating substantially more total trade volume.
- Confidence intervals (error bars) are notably wider for Fear and Extreme Greed, reflecting greater variability in trade sizes — consistent with a more polarised trader base (some very large, some very small positions) in extreme sentiment conditions.

Key Insight: *Traders in extreme sentiment states (Extreme Greed and Fear) trade with larger average positions, but also with more variability. This elevated uncertainty in trade sizing may contribute to the wider PnL distributions seen in Figure 1.*

Figure 4: PnL by Sentiment & Trader Consistency

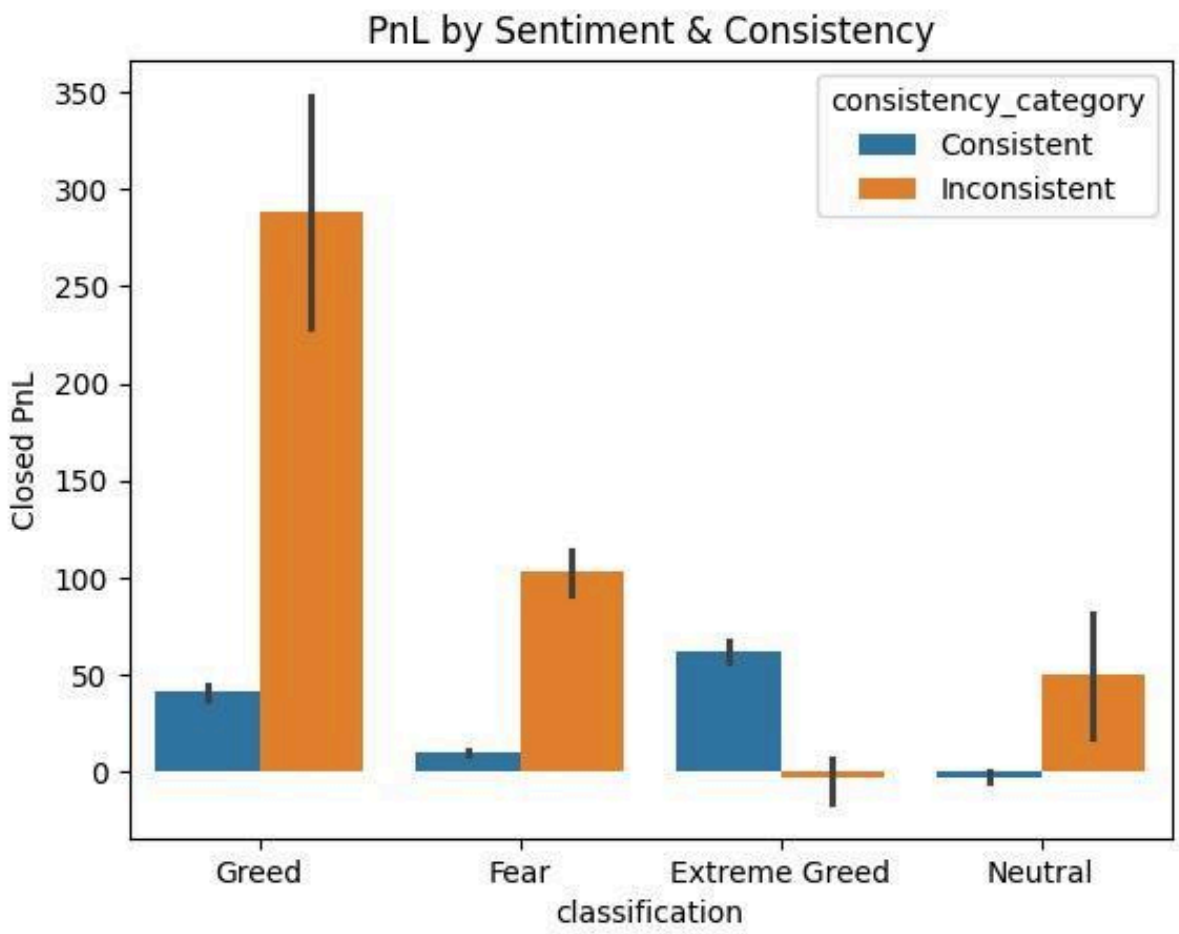

Figure 4 — Grouped bar chart comparing average Closed PnL for Consistent vs. Inconsistent traders across all sentiment classifications.

Interpretation

This chart is one of the most counterintuitive findings of the entire analysis, challenging conventional wisdom that consistent trading strategies outperform variable ones:

- Inconsistent traders (orange bars) dramatically outperform Consistent traders (blue bars) in three of the four sentiment categories — Greed (288 vs 41), Fear (103 vs 10), and Neutral (50 vs -2).

- The Greed premium is particularly striking: Inconsistent traders earn nearly 7× the PnL of Consistent traders in bullish markets, suggesting that adaptable or opportunistic strategies are rewarded when markets are trending upward.
- Extreme Greed is the sole exception: Consistent traders modestly outperform Inconsistent ones (62 vs -3), hinting that disciplined risk management becomes valuable when euphoria peaks and reversal risk is highest.
- Consistent traders show near-zero or negative PnL in Neutral and Fear environments, suggesting their strategies may be optimised for specific conditions and struggle in uncertainty.
- Error bars are substantially wider for Inconsistent traders, confirming that their higher average returns come with higher variance — a classic risk-return tradeoff.

Key Insight: *Inconsistency in trading — often considered a flaw — appears to be a feature in trending markets. Traders who adapt their approach to changing conditions, rather than applying a fixed strategy, may capture more alpha. However, they also carry higher downside risk, as reflected in the wider confidence intervals.*

Detailed Key Findings

1. Sentiment and Profit & Loss

Market sentiment shows a clear relationship with average trader PnL, though variability within sentiment categories is high:

- Greed periods yield the highest average Closed PnL (87.89), nearly double that of Fear (50.05).
- Extreme Greed periods show a surprisingly modest average PnL (25.42), slightly above Neutral (22.23).
- Wide PnL distributions across all sentiments confirm that sentiment alone does not guarantee profitability.

2. Win Rate by Sentiment

Win rates (proportion of profitable trades) differ meaningfully across sentiment states:

- Extreme Greed: 49.01% win rate — traders come closest to 50/50 profitability.
- Fear: 41.51% win rate — roughly 4 in 10 trades are profitable during fearful markets.
- Neutral: 31.72% — the lowest win rate, suggesting directionless markets are hardest to profit from.

3. Trader Behavior by Sentiment

Market sentiment affects not just outcomes but trading behavior patterns:

- Trade volume peaks sharply during Fear (133,871 trades), suggesting panic-driven or opportunistic activity.
- Average trade size is largest during Extreme Greed ($5,660), indicating traders commit more capital during bullish extremes.
- Buy/sell balance tilts slightly toward selling in Greed (57.5% sell) and Extreme Greed (51.6% sell), consistent with profit-taking.

4. Segment-Based Insights

High vs. Low Trade Size

- High-size traders consistently achieve significantly higher average PnL in all sentiment conditions.
- During Greed, high-size traders earn an average PnL of 175.72 vs. just 12.59 for low-size traders — a 14× difference.

Consistent vs. Inconsistent Traders

- Inconsistent traders outperform in Fear (102.96 vs 10.40), Greed (288.16 vs 41.07), and Neutral (50.56 vs -2.39).
- Consistent traders perform better only in Extreme Greed (61.77 vs -2.77).

Frequent vs. Infrequent Traders

- Infrequent traders show substantially higher PnL in Fear (75.64 vs 46.01), Greed (234.89 vs 70.58), and Neutral (356.72 vs 4.06).
- Only in Extreme Greed do infrequent traders show negative PnL.

5. Predictive Modelling

A Random Forest Classifier was built to predict whether a trader would be profitable the following day, using daily-level features including sentiment encoding, trade count, average size, and win rate.

- Model accuracy: approximately 64.29% — outperforming a random baseline of 50%.
- Market sentiment and trader behaviour features carry a meaningful predictive signal, even in noisy financial markets.
- Moderate accuracy is consistent with the difficulty of predicting short-term market outcomes and should be interpreted as directional rather than prescriptive.

Conclusion

Market sentiment has a demonstrable influence on cryptocurrency trader performance, trade volume, and trade sizing. While 'Greed' conditions yield the highest average PnL, the data reveals that segment membership — particularly trade size and trading frequency — is a far stronger determinant of profitability than sentiment alone.

The most striking finding is that infrequent and inconsistent traders tend to outperform their frequent and consistent peers under most sentiment conditions, suggesting that selective, high-conviction trading strategies generate superior returns in volatile crypto markets.

The predictive model's 64.29% accuracy provides encouraging evidence that sentiment-augmented features can inform profitability forecasting, opening pathways for further feature engineering and model refinement.